

STORNOWAY, United Kingdom

WMO: 030260

Lat: 58.2136N

Lon: 6.3189W

Elev: 26

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.9	31.9	23.5	17.8	34.5	25.5	19.6	35.3	43.8	44.1	40.8	44.0	7.0	310	0.787

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	65.9	59.9	63.1	58.2	61.3	57.1	61.1	64.3	59.4	61.9	58.1	60.4	10.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	76.1	62.7	58.1	72.3	60.8	56.8	68.9	59.4	27.2	64.6	26.0	62.1	25.2	60.5	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.7	32.2	28.7	DB	24.6	71.6	5.1	3.5	20.9	74.1	17.9	76.2	15.1	78.2	11.3	80.7	(4)
(5)			WB	23.9	64.1	3.0	2.8	21.7	66.1	19.9	67.8	18.2	69.3	16.0	71.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	41.2	41.0	42.3	45.2	48.6	52.8	56.1	56.3	53.6	49.3	44.7	41.7		
	DBStd	6.90	4.14	4.29	4.08	4.16	4.13	3.46	3.37	3.05	3.48	4.13	4.60	5.09		
	HDD50	1498	274	253	239	149	75	11	0	9	64	164	259			
	HDD65	6290	739	673	704	593	508	366	276	271	343	487	608	722		
	CDD50	685	0	1	1	7	32	95	190	194	116	41	6	2		
	CDD65	1	0	0	0	0	0	0	0	1	0	0	0	0		
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	14.1	16.4	16.4	15.9	13.8	12.8	12.7	11.4	11.7	12.9	14.6	14.8	15.5		
	PrecAvg	46.9	5.2	3.8	4.0	2.8	2.5	2.5	2.9	3.3	4.3	5.2	5.1	5.1		
(15) Precipitation	PrecMax	60.9	10.6	10.7	8.1	4.6	6.0	4.4	6.5	8.0	8.1	9.1	9.0	10.6		
	PrecMin	31.5	1.0	0.4	0.7	0.0	0.5	0.6	0.9	0.6	1.3	1.5	1.5	0.9		
	PrecStd	6.4	2.1	2.2	1.7	1.0	1.2	0.9	1.1	1.6	1.6	1.9	1.7	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	52.6	53.3	59.5	64.2	66.8	70.2	69.2	64.6	59.9	56.4	54.1		
		MCWB	50.5	50.1	48.8	51.8	56.7	59.3	63.3	62.3	58.1	55.9	53.6	51.9		
	2%	DB	50.5	50.9	51.2	55.0	59.4	62.0	66.1	64.9	61.5	57.9	54.2	52.5		
		MCWB	48.9	49.1	48.3	49.6	53.5	56.9	60.7	59.8	57.1	54.7	51.9	50.7		
	5%	DB	49.3	49.4	49.7	52.8	56.7	59.5	63.3	62.4	60.0	56.6	52.9	51.2		
		MCWB	47.7	47.7	47.2	48.6	51.8	55.0	58.8	58.2	56.4	53.9	50.7	49.4		
	10%	DB	47.8	47.8	48.4	51.1	54.7	57.8	61.1	60.8	58.6	55.4	51.7	49.7		
		MCWB	46.0	45.9	46.2	47.7	50.6	53.8	57.3	57.1	55.6	52.8	49.6	47.9		
	0.4%	WB	51.1	51.0	50.9	52.5	57.5	60.3	64.2	63.2	60.0	56.9	54.5	52.9		
		MCDB	51.6	51.7	52.4	56.9	63.2	65.4	69.0	68.1	62.0	58.4	55.5	53.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.4	49.5	49.1	50.8	54.6	57.6	61.6	60.9	58.4	55.7	52.7	51.2		
		MCDB	50.2	50.4	50.3	53.8	58.2	60.9	64.8	63.7	60.2	57.2	53.8	52.1		
	5%	WB	48.0	48.1	47.8	49.5	52.7	55.9	59.7	59.2	57.4	54.6	51.2	49.7		
		MCDB	49.1	49.2	49.2	51.9	55.7	58.4	62.4	61.4	59.2	56.2	52.5	50.9		
	10%	WB	46.3	46.1	46.5	48.4	51.2	54.7	58.0	58.0	56.3	53.3	49.7	48.0		
		MCDB	47.7	47.6	48.2	50.7	54.0	57.0	60.4	60.0	58.2	55.0	51.5	49.7		
	5% DB	MDBR	7.3	8.0	8.7	9.7	10.2	8.9	9.0	8.9	8.8	8.4	8.0	8.1		
		MCDBR	8.1	8.1	9.1	11.3	12.7	11.3	12.1	10.7	9.8	8.4	8.0	8.5		
		MCWBR	8.2	7.7	7.6	7.4	8.0	7.2	7.5	6.9	6.9	7.0	8.0	8.9		
		MCWBR	8.3	8.0	8.5	10.2	11.9	10.3	11.2	9.7	8.6	8.0	8.4	8.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.7	8.0	7.9	7.4	8.2	7.2	7.5	7.1	6.9	7.6	8.8	9.4		
	taub		0.353	0.356	0.370	0.383	0.382	0.373	0.377	0.369	0.365	0.350	0.329	0.320		
		taud	2.109	2.240	2.298	2.318	2.378	2.444	2.446	2.477	2.452	2.360	2.204	2.100		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		139	208	243	261	270	274	270	265	246	210	150	112		
		Edh at Noon	18	24	30	34	35	33	32	29	26	21	16	13		
	RadAvg		104	289	644	1132	1508	1524	1408	1136	744	378	144	69		
		RadStd	12	28	45	76	109	137	99	74	37	33	13	8		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	+1.82	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page